

Chapter 3: The chain of infection

Objectives

- Understand the chain of infection.
- Recognize the epidemiological importance of human carriers

As described before, the traditional epidemiologic triad model holds that infectious diseases result from the interaction of agent, host, and environment. More specifically, transmission occurs when the agent leaves its reservoir or host through a portal of exit, is conveyed by some mode of transmission, and enters through an appropriate portal of entry to infect a susceptible host. This sequence is sometimes called the chain of infection.

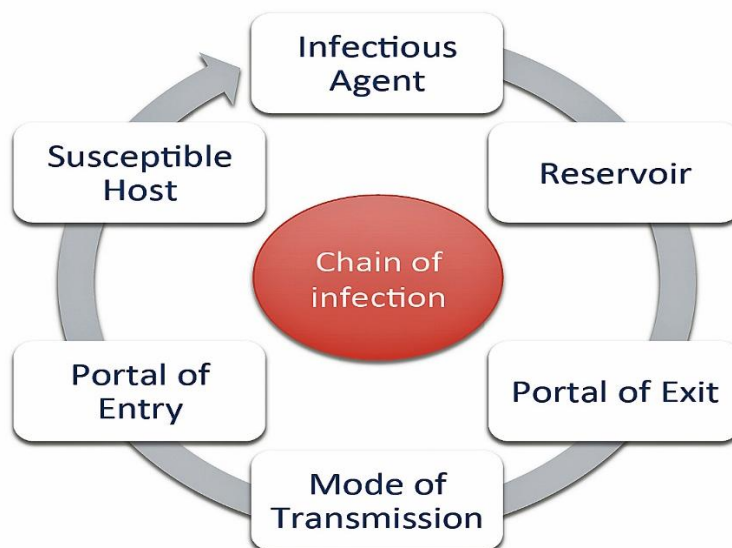


Fig. (3) : The chain of infection

Reservoir

The reservoir of an infectious agent is the habitat in which the agent normally lives, grows, and multiplies. Reservoirs include humans, animals, and the environment. The reservoir may or may not be the source from which an agent is transferred to a host. For example, the reservoir of *Clostridium botulinum* is soil, but the source of most botulism infections is improperly canned food containing *C. botulinum* spores.

Human reservoirs. Many common infectious diseases have human reservoirs. Diseases that are transmitted from person to person without intermediaries include the sexually transmitted diseases, measles, mumps, streptococcal infection, and many respiratory pathogens. Because humans were the only reservoir for the smallpox virus, naturally occurring smallpox was eradicated after the last human case was identified and isolated

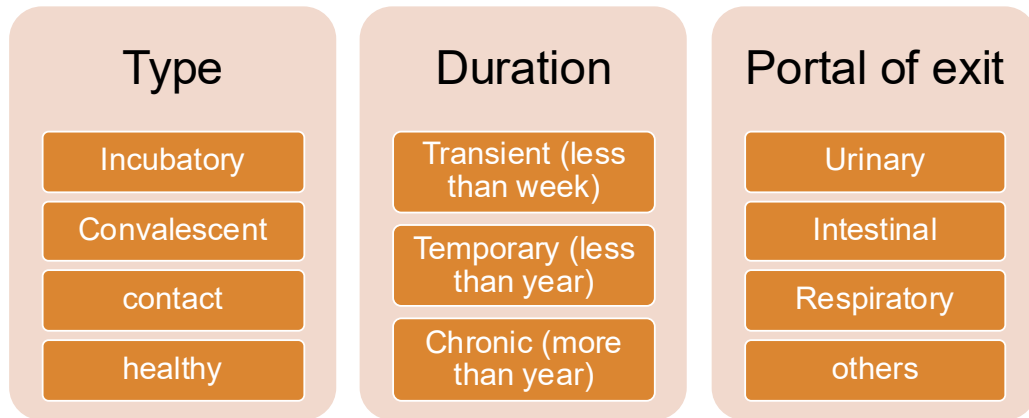
Human reservoirs include:

- Cases: patients show manifestation of the disease.
- Carriers: a carrier is apparently healthy individual who is **infected and harbors the pathogenic organism as a foci of infection** in different parts of his body. *Fig (4)*.

Carrier has an epidemiological importance

- 9- He doesn't show manifestations of disease, but he can spread infection.
- 10- pass free in the community
- 11- diagnosis needs laboratory investigations
- 12- the danger of carrier in some occupations dealing with community members: food handlers, nurses, teachers.

Carriers



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Fig. (4): Human carriers

- Incubatory carrier: some cases is infectious in its incubation period.
- Convalescent carrier : a certain percent of convalescents (recovered) cases continues to harbor the causative organism in particular foci for varied period of time (weeks, months, years or lifelong)
- Contact carrier: contacts of cases or carriers of some infectious diseases may get infected but not diseased. Infection is eliminated in around two weeks.
- Healthy carrier: inhabitants of endemic areas of infectious diseases may get infected from polluted environment; water supply and food. Infection is eliminated in two weeks.

Animal reservoirs. Humans are also subject to diseases that have animal reservoirs. Many of these diseases are transmitted from animal to animal, with

humans as incidental hosts. The term zoonosis refers to an infectious disease that is transmissible under natural conditions from vertebrate animals to humans. Long recognized zoonotic diseases include brucellosis (cows and pigs), anthrax (sheep), plague (rodents), trichinellosis/trichinosis (swine), tularemia (rabbits), and rabies (bats, raccoons, dogs, and other mammals).

Environmental reservoirs. Plants, soil, and water in the environment are also reservoirs for some infectious agents. Many fungal agents, such as those that cause histoplasmosis, live and multiply in the soil. Outbreaks of Legionnaires disease are often traced to water supplies in cooling towers and evaporative condensers, reservoirs for the causative organism *Legionella pneumophila*.

Portal of exit

Portal of exit is the path by which a pathogen leaves its host. The portal of exit usually corresponds to the site where the pathogen is localized. For example, influenza viruses and *Mycobacterium tuberculosis* exit the respiratory tract, schistosomes through urine, cholera vibrios in feces, *Sarcoptes scabiei* in scabies skin lesions, and enterovirus 70, a cause of hemorrhagic conjunctivitis, in conjunctival secretions. Some bloodborne agents can exit by crossing the placenta from mother to fetus (rubella, syphilis, toxoplasmosis), while others exit through cuts or needles in the skin (hepatitis B) or blood-sucking arthropods (malaria). Fig (5).



Fig. (5): Exit of infection

Modes of transmission

An infectious agent may be transmitted from its natural reservoir to a susceptible host in different ways. There are different classifications for modes of transmission. Here is one classification:

1. Direct
 - a. Direct contact (skin-to-skin contact, kissing, and sexual intercourse)
 - b. Droplet spread (fig. 6)
2. Indirect
 - a) Airborne (fig. 7)
 - b) Vehicle borne (food, water, biologic products (blood), and fomites (inanimate objects such as handkerchiefs, bedding, or surgical scalpels). A vehicle may passively carry a pathogen as food or water may carry hepatitis A virus. Alternatively, the vehicle may provide an environment in

which the agent grows, multiplies, or produces toxin as improperly canned foods provide an environment that supports production of botulinum toxin by *Clostridium botulinum*.

- c) Vector borne. Vectors such as mosquitoes, fleas, and ticks may carry an infectious agent through purely mechanical means or may support growth or changes in the agent. Examples of mechanical transmission are flies carrying *Shigella* on their appendages and fleas carrying *Yersinia pestis*, the causative agent of plague, in their gut. In contrast, in biologic transmission, the causative agent of malaria or guinea worm disease undergoes maturation in an intermediate host before it can be transmitted to humans

Another classification of modes of transmission of infection

1. Droplet infection

- a) Droplet transmission (fig. 6)
- b) Airborne transmission (fig. 7)
- c) Dust: big droplets fall on the floor, get dried including the organisms and rise again in air with dust
- d) Milk : contaminated by 2 respiratory pathogens and invade upper respiratory mucosa : haemolytic *Streptococci* and diphtheria bacilli.

Droplet transmission

Characteristics of droplet particles

- **Size:** large particles (> 5 microns)
- Note: One droplet contains many viruses
- **Behavior:** succumb to gravity, meaning they fall down after travelling in the air
 - **Travel Distance:** short distances, usually land within 1 to 2 meters.



T-zone

Transmission process

- Patient can transmit infection by coughing, sneezing
- We can get infected when droplets contact directly or indirectly mucous membranes of our eyes, nose, or mouth (T-zone).

Fig. 6: Droplet transmission.

Airborne transmission

Characteristics of aerosols particles

- **Size:** tiny particles (≤ 5 microns)
 - **Behavior:** remain active in the air over time and distance
 - **Travel Distance:** can spread over long distances, depending of initial velocity, airflow, and humidity.
- Note: infected person does not need to be in the same room than others to infect them.



Respirator

Transmission process

- Patient can transmit infection by coughing, sneezing, talking, exhaling or when an Aerosol Generated Procedure (AGP) is performed
- We can get infected when inhaling these tiny infected particles. Hence the importance of using a fit-tested respirator.

Fig. 7: Airborne transmission.

2- Ingestion (feco-oral transmission):

by contaminated hands, food or water (fig. 8).

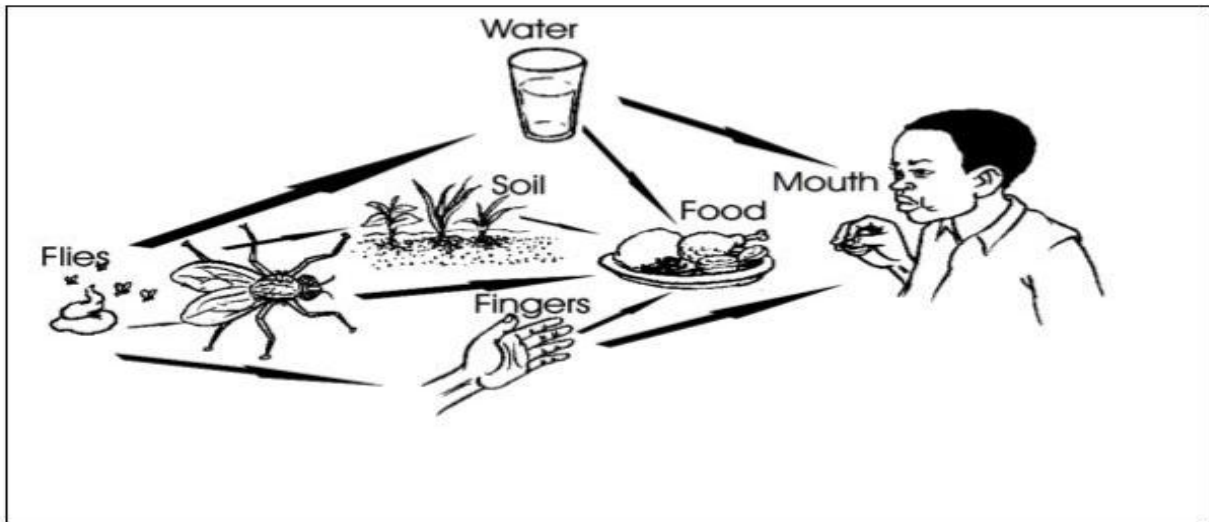


Fig. (8): Feco-oral transmission

3- arthropod- borne infections: mosquitos, fleas , cockroaches.

4- contact infection: invasion of skin and mucus membrane by pathogenic organisms.

5- blood: HBV , HCV, HIV, Malaria

6- vertical : from mother to infant (during pregnancy labor or breast feeding)

Portal of entry

The portal of entry refers to the manner in which a pathogen enters a susceptible host. The portal of entry must provide access to tissues in which the pathogen can multiply or a toxin can act. Often, infectious agents use the same portal to enter a new host that they used to exit the source host. For example, influenza virus exits the respiratory tract of the source host and enters the respiratory tract of the new host. In contrast, many pathogens that cause gastroenteritis follow a so-called “fecal-oral” route because they exit the source host in feces, are carried on inadequately washed hands to a vehicle such as food, water, or utensil, and enter a new host through the mouth.

Exposed host

The final link in the chain of infection is a susceptible host. Susceptibility of a host depends on genetic or constitutional factors, specific immunity, and nonspecific factors that affect an individual’s ability to resist infection or to limit pathogenicity. An individual’s genetic makeup may either increase or decrease susceptibility. For example, persons with sickle cell trait seem to be at least partially protected from a particular type of malaria. Specific immunity refers to protective antibodies that are directed against a specific agent. Such antibodies may develop in response to infection, vaccine, or toxoid (toxin that has been deactivated but retains its capacity to stimulate production of toxin antibodies) or may be acquired by transplacental transfer from mother to fetus or by injection of antitoxin or immune globulin. Nonspecific factors that defend against infection include the skin, mucous membranes, gastric acidity, cilia in the respiratory tract, the cough reflex, and

nonspecific immune response. Factors that may increase susceptibility to infection by disrupting host defenses include malnutrition, alcoholism, and disease or therapy that impairs the nonspecific immune response.

Knowledge of the chain of infection provides a basis for determining appropriate control measures.

Control measures are directed at:

- Controlling or eliminating agent at source of transmission (interrupt vector borne transmission, measures may be directed toward controlling the vector population, such as spraying to reduce the mosquito population)
- Protecting portals of entry (bed nets are used to protect sleeping persons from being bitten by mosquitoes that may transmit malaria.)
- Increasing host's defenses (Vaccinations promote development of specific antibodies that protect against infection. On the other hand, prophylactic use of antimalarial drugs, recommended for visitors to malaria-endemic areas, does not prevent exposure through mosquito bites, but does prevent infection from taking root)